SIVA S

AI & Data Engineering | Python Backend | ETL & Automation | LLM Integrations

Tenkasi, Tamil Nadu, India | +91 8072861947 | sivavpy2005@gmail.com

linkedin.com/in/siva-sankaravel-400145323 | github.com/Siva-Sankaravel | siva-sankaravel.github.io/Portfolio

PROFESSIONAL SUMMARY

AI & Data Science graduate (B.Tech, 2026) and Data Engineer Trainee with hands-on experience building Python backend services, ETL pipelines, REST API integrations, and AI-powered workflow automation. Worked on PostgreSQL-backed content operations, MCP-based AI agent tool integrations, WhatsApp messaging services, and LLM-assisted content publishing workflows. Strong foundation in SQL, data validation, machine learning, Power BI, and cloud infrastructure. Seeking Data Engineer, Python Backend, or AI Automation roles to build reliable, data-driven solutions.

TECHNICAL SKILLS

Programming Languages: Python, SQL, HTML/CSS, JavaScript (basic), PowerShell

Databases: PostgreSQL, MySQL, SQLite, SQL Query Writing, Upserts, Schema Design

Backend & APIs: Flask, FastAPI, REST API Development, Swagger/OpenAPI, HTTPX, Requests, JWT Authentication

Data Engineering: ETL Pipelines, Data Ingestion, Data Cleaning, Data Validation, Batch Processing, Pandas, NumPy, openpyxl

AI & Automation: LLM Integration, Prompt Engineering, Claude AI, Model Context Protocol (MCP), FastMCP, n8n, APScheduler, Workflow Automation, Web Scraping (Selenium, Playwright, BeautifulSoup)

Machine Learning & Analytics: Scikit-learn, Random Forest, K-Means Clustering, Feature Engineering, EDA, Power BI, Tableau, Matplotlib, Seaborn, Microsoft Excel

Cloud & DevOps: AWS, Terraform, Docker, GitHub Actions, CI/CD

Tools: Git, GitHub, Postman, VS Code

PROFESSIONAL EXPERIENCE

Data Engineer Trainee Jun 2026 – Nov 2026

MercuryMinds Technologies, Chennai

- Developed major components of Python data engineering and automation workflows covering data ingestion, relational databases (PostgreSQL, MySQL), REST services, and third-party API integrations.
- Contributed to modular Flask backend services for content lifecycle management, including API handlers, data validation, database interactions, and scheduled background processing with APScheduler.
- Implemented data ingestion and transformation logic for Excel/CSV sources, including multi-sheet parsing, header and field normalization, duplicate handling, and row-level error reporting.
- Built AI-assisted automation and API-based messaging workflows connecting LLM-generated content, campaign scheduling, and external platforms, with execution tracking and recoverable failure handling.
- Built web scraping workflows using Selenium, Playwright, and Requests to collect and analyze market data; conducted AI-assisted market analysis to map customer segments, product adoption, and industry trends.

Data Science Intern 2024

Live Stream Technologies, Coimbatore

- Collected, cleaned, and preprocessed large datasets to improve data quality and accuracy for analysis and machine learning use cases.
- Developed machine learning models for prediction and created dashboards in Tableau, Power BI, and Matplotlib to communicate trends and business metrics to stakeholders.
- Collaborated with cross-functional teams to analyze trends and optimize business strategies using statistical techniques.

INDUSTRY PROJECTS

Content Operations & Data Pipeline Backend *Python, Flask, PostgreSQL, Polars, REST APIs*

- Developed backend components for a content operations platform supporting data ingestion, content metadata management, workflow tracking, and CMS integration.
- Implemented spreadsheet ingestion, XML sitemap parsing, field normalization, relational data upserts (PostgreSQL `ON CONFLICT`), and validation/error logging for repeatable, idempotent processing.
- Contributed to editorial review and publishing workflows with scheduled tasks, execution status tracking, and access-controlled API interactions.

AI Agent Tool Integration Gateway (MCP) *Python, FastMCP, HTTPX, REST APIs, JWT*

- Built components of a Model Context Protocol (MCP) integration layer exposing structured tools that let AI assistants interact with existing backend services through REST interfaces.
- Implemented request forwarding, payload normalization, authentication-context propagation, and structured tool responses while keeping business rules inside the backend services.

WhatsApp Messaging Automation Integration *Python, Requests, Kapso API, Meta WhatsApp Cloud API*

- Integrated a WhatsApp messaging service with scheduled campaign workflows, supporting approved template messages for outreach and session-based text replies within the 24-hour window.
- Added E.164 phone-number normalization, environment-based credential configuration (`python-dotenv`), HTTP timeout handling, and structured delivery results that isolate individual failures from batch runs.

AI-Assisted Content Generation & Publishing Pipeline *Python, MySQL, openpyxl, Claude AI, WordPress REST API*

- Developed an ETL pipeline combining Excel queues and MySQL data with LLM-assisted (Claude AI) generation of structured web content, FAQs, metadata, Open Graph tags, and JSON-LD schema markup.
- Integrated generated content with the WordPress REST API for draft creation; added duplicate detection, taxonomy validation, HTML sanitization, input validation, and execution logs.
- Implemented checkpoint-based processing, rate-limit backoff, and retry/error-handling logic to recover from interrupted batch operations.

PERSONAL & ACADEMIC PROJECTS

Customer Churn Prediction System *Python, Scikit-learn, Random Forest, FastAPI, Render*

- Built a customer churn classification workflow using Random Forest with preprocessing and model evaluation, exposing probability-based predictions through a FastAPI backend deployed on Render.
- Developed a web frontend (HTML/CSS/JavaScript) on GitHub Pages that consumes the prediction API in real time.

AI Sales Forecasting Dashboard *Python, Pandas, NumPy, Scikit-learn, Power BI*

- Developed a revenue forecasting workflow using feature engineering, trend analysis, and predictive modeling.
- Designed an interactive Power BI dashboard for forecasted sales, KPI trends, and region-wise insights.

Customer Segmentation Analytics *Python, K-Means, Power BI*

- Performed exploratory data analysis, feature scaling, outlier detection, and K-Means clustering to group customers by behavior and spending.
- Built Power BI and Python visualizations of segments with recommendations for targeted marketing.

AWS Infrastructure Automation *Terraform, AWS, GitHub Actions*

- Defined AWS infrastructure as code using Terraform, covering VPC networking, EC2, Application Load Balancer, Auto Scaling, RDS, and S3.
- Connected infrastructure provisioning workflows to a GitHub Actions CI/CD pipeline.

Smart Umbrella for Two-Wheelers *Arduino, IoT Sensors, Motor Control*

- Developed a sensor-driven motorized umbrella prototype for two-wheelers using water, wind, and temperature sensors; recognized as a finalist in MSME Hackathon 2024.

EDUCATION

B.Tech in Artificial Intelligence & Data Science 2022 – 2026
P.A. College of Engineering and Technology, Pollachi | CGPA: 7.42

Higher Secondary (Class XII) 2020 – 2022
SMA Matric Higher Secondary School, Tenkasi | 83%

Secondary School (Class X) 2019 – 2020
SMA Matric Higher Secondary School, Tenkasi | 92%

WORKSHOPS & ACTIVITIES

- Workshop: Artificial Intelligence, IIT Madras Research Park
- Webinar: Prompt Engineering – End User, Koenig Step Forward
- Leadership: Event Coordinator, AIVORA, PACET; Member, Instagram Engineers Club, PACET